Challenge Track 1: Predicting Border Dispute Risk Using AI & Geospatial Data

Monitoring the Situation: Anomaly Detection at the Thailand-Cambodia Border Using Attention-Based LSTMs

Nitya Kumar, Siya Brahmbhatt, Savannah Gill, Noah Orejuela, Jared Yili

Rutgers University–New Brunswick

16 March 2026

Abstract

The Thailand-Cambodia border dispute represents one of Southeast Asia's most enduring territorial conflicts. Violence and unrest sporadically erupts around disputed areas, presenting the challenge for analysts and decision-makers of anticipating and planning for such volatility. We look to the field of conflict forecasting, which examines trends across multiple dimensions, including political developments, geographic clustering, temporal conflict patterns, and shifts in media narratives to predict and detect violence and unrest. Here, we describe an attention-based long short-term memory (LSTM) neural network approach to anomaly detection, specifically designed for state-based conflict at the Thailand-Cambodia border. We build on the current body of literature and experiment with separate spatial and temporal attention mechanisms. Our results indicate that escalation risk remains highly concentrated around historically contested sites even during periods of heightened political or informational tension. We also demonstrate how artificial intelligence (AI)-driven geospatial modeling can contribute to early-warning systems for interstate border conflicts and support more proactive conflict monitoring in Southeast Asia.

Keywords*: Thailand, Cambodia, anomaly detection, long short-term memory (LSTM)*

Background

The Thailand-Cambodia conflict's escalation to direct armed confrontation must be understood in adjunction with its historical continuum of incomplete institutionalization and fight over natural resources. Border incidents through early 2026 have remained largely confined to a set of disputed zones across the mainland border, specifically spurred by a religious attachment to Hindu temples located on the east and west sides of the international border. Quantifiably, Preah Vihear Temple and Ta Muen Thom Temple exhibit a disproportionately high probability of militarized incidents, as the theatre of conflict remains centered around 25 miles of these sites. Despite the geographically limited scope of past clashes, however, the broader security escalation of the dispute has continued to intensify.

The Thailand and Cambodia 2001 Memorandum of Understanding (MOU-44) established a Joint Boundary Commission and acknowledged additional overlapping land and maritime claims, including extended contested zones within the Gulf of Thailand near Koh Kood. These high-risk offshore Overlapping Claim Areas (OCA) make up a 26,500 km² maritime zone reported to contain 10–11 trillion cubic feet of natural gas and around 300 million barrels of crude oil (Chalermpalanupap, 2025). As a result, both Thailand and Cambodia have regarded control over these zones as a significant strategic advantage, while historically making periodic advancements in an effort to secure long-term energy resources and maritime jurisdiction which have further bled into intensifying mainland conflicts. It is important to note, however, that skirmishes near Koh Kood have an unlikely probability of repeated occurrence due to the multi-country claim over the Gulf of Thailand, as neither Thailand nor Cambodia wish to draw in other ASEAN countries (Malaysia and Vietnam) into the ongoing conflict.

As of July 2025, open-source intelligence (OSINT) measures Thailand's capital defense budget as approximately $5.75 billion USD, significantly exceeding the $1.3 billion allocation of Cambodia (SIPRI). While this disparity reflects Thailand's greater conventional military capacity, both states continue to sustain military deployments in contested zones, increasing the probability of direct armed confrontation in those zones and along the greater shared-border. These repeated confrontations, however, have historically manifested as a low-intensity centralized conflict characterized by intermittent and geographically localized clashes, as opposed to sustained total warfare. Exchanges of fire since 2008 have frequently resulted from patrols encountering one another in poorly demarcated areas, where both sides maintain a strong troop presence in order to reinforce their respective territorial claims (Wagener, 2011). As confrontations extended beyond brief skirmishes into multi-day engagements, as evident during the early 2011 clashes, such incidents are unlikely to be accidental and instead suggest political tolerance or tacit authorization from national leadership seeking to signal resolve while avoiding escalation into full-scale war.

This pattern of sporadic escalation presents a significant challenge for forecasting conflict. Low-intensity border disputes often oscillate between prolonged periods of relative calm and sudden bursts of violence driven by shifts in domestic politics, strategic signaling, or perceived windows of opportunity. Such dynamics complicate traditional conflict prediction frameworks that rely primarily on structural indicators or casualty trends (Chadefaux, 2017). Consequently, modern research has increasingly turned toward computational and data-driven approaches for detecting early signals of escalation. Current works have discussed existing statistical methodologies for forecasting political violence and identifying anomalies within conflict data, including graph or network-based approaches, hidden Markov models, and

tree-based algorithms (Schrodt et al., 2013). These approaches provide a foundation for integrating political, geographic, and temporal indicators into predictive systems capable of identifying abnormal activity patterns that may precede localized conflict escalation, but recent advancements in machine learning architecture have allowed for more complex analyses.

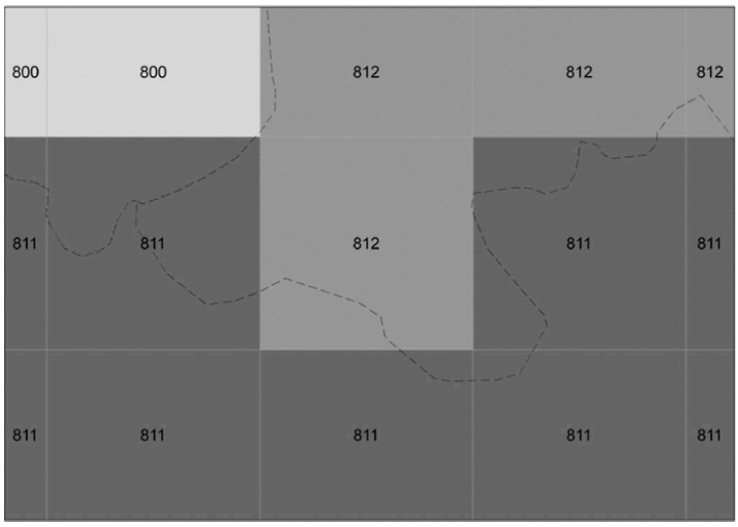
Methods

We use two primary open-source datasets: the Armed Conflict Location and Event Data Project (ACLED) and the Global Database of Events, Language, and Tone (GDELT). ACLED provides structured records of political violence and protest events, including information such as event type, location coordinates, actors involved, and fatalities (Raleigh et al., 2010). This dataset allows us to track the spatial and temporal distribution of state-based conflict events near the Thailand-Cambodia border. We exclude the "strategic developments" event type.

GDELT aggregates global news reporting and converts it into structured event data, capturing mentions of political actors, locations, and the tone of media coverage. Unlike ACLED, which records confirmed conflict events, GDELT reflects shifts in media attention and rhetoric that may signal rising tensions before violence occurs (Macis et al., 2024). Together, these datasets allow the model to incorporate both observed conflict activity and informational signals when analyzing escalation risk.

Current works (such as by Hegre et al. (2019)) analyze events at a country and/or subnational level. The former level does not provide enough precision to dissect an international, localized border dispute, and there are seemingly infinite ways to implement the latter. Tollefsen et al. (2012) at the Peace Research Institute Oslo (PRIO) attempt to address the inconsistency of forecasting subdivisions with their PRIO-GRID standard. It divides the world into 0.5 x 0.5

decimal degree (approximately 55 x 55 km at the equator) rectangular cells, a size which is small enough to give smaller nations multiple cells and not too large as to increase the size and complexity of datasets and thus the cost of training a model. We use PRIO-GRID cells for our spatial unit; our temporal unit is in days, leading to a combined spatiotemporal unit of cell-days.



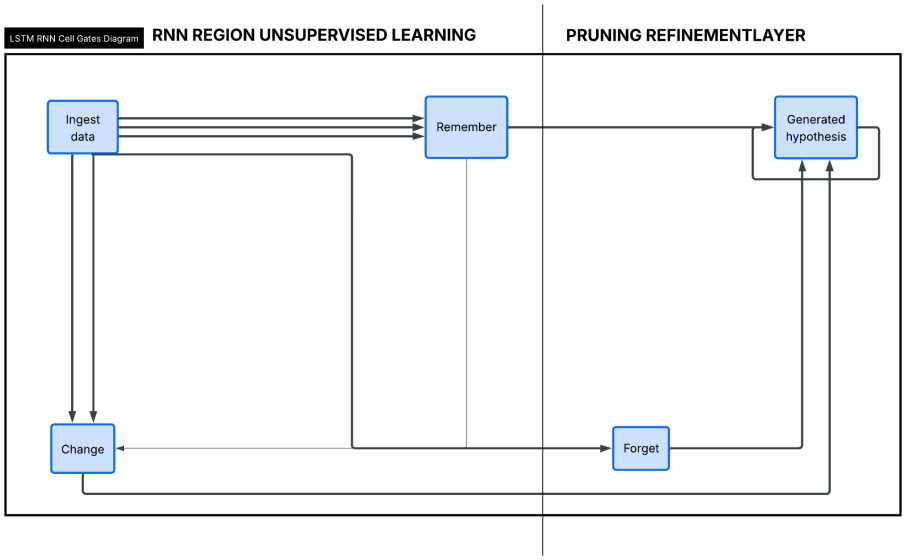
PRIO-GRID cells at the border between Thailand, Cambodia, and Laos, with grayscale coloring indicating which color the cells belong to (Tollefsen et al., 2012).

We refrain from becoming mired in the technical details of our model approach. Nevertheless, there are some concepts to understand before we discuss it. We start with an artificial neural network, which started off based on the synaptic model of the brain. In this simplification of the brain, signals travel between neuron cells. The synapses between neurons change depending on the signal they carry, influencing the signals that reach other neurons. In an artificial neural network, neurons apply a mathematical function to an input and pass it on to another layer of neurons, eventually reaching an output. To apply a neural network to ‘learn’ a pattern and apply it in problems like text recognition, a model’s parameters (weights) are iteratively adjusted to minimize the difference (error) between its output and the desired output.

Training data must be provided for the model to learn the patterns of the desired output. In other words, this process of learning is *supervised*.

A recurrent neural network (RNN) allows sequential outputs of neurons to flow both forwards and backwards across layers; receiving neurons process each input according to their order. This order-based design makes RNNs well-suited to capturing sequential and temporal patterns. However, RNNs suffer vanishing and exploding gradient problems wherein error signals are cumulatively shrunk or amplified to the point where they are no longer useful for training. Long short-term memory (LSTM) is a type of RNN that addresses the vanishing gradient problem using a cell and specialized forget, remember, and change "gates" (Hochreiter and Schmidhuber, 1997). The cell retains information over time and the gates control how information is discarded, stored, or transformed, enabling LSTMs to remember patterns over much longer periods of time.

Building on the work of Macis et al. (2024), we approach our given problem using a LSTM autoencoder for anomaly detection. An autoencoder enables a reconstruction-based method: an encoder compresses data into a representation of the input state, and a decoder uses that representation to reconstruct the data as accurately as possible. Rather than attempting to train a model to predict a variable using defined independent and dependent variables–that is, *supervised* learning–in an inherently volatile, sparse, and low-intensity conflict, we can train our LSTM autoencoder to reconstruct what 'normal' days look like. This strategy takes the form of *unsupervised* learning, reducing our reliance on labeled data. As conflict precursors occur in the data stream, the normal state reconstruction will worsen in performance and we can use an error metric as an anomaly score.



Simplified architecture of an LSTM module.

The LSTM autoencoder forms the foundation of our approach. We next experiment with using attention mechanisms. Attention is a method that weights different parts of sequential data based on their relevance to the current prediction; to accomplish this, an attention mechanism can select relevant states from multiple previous time steps. This approach allows our model to learn which factors are most influential in reconstructing the normal state and adapt as conditions in the input change (Qin et al., 2017). When applied to our dataset—a mosaic of geopolitical news and conflict event reports—this mechanism allows the network to retain and emphasize signals from relevant contextual conditions.

Drawing from Yuxuan et al. (2018), we expand on Macis et al. (2024) using global spatial and temporal attention mechanisms. The global spatial module in the encoder correlates cells in proximity to each other. The temporal module is in the decoder, adaptively selecting states from the encoder to produce a predictive output sequence. This model architecture allows us to account for changes in data density and content depending on geopolitical events in separate temporal and spatial cells.

Results

We began our ingest on ACLED and GDELT datasets and ran into two separate problems quickly. Both datasets carry different identifiers for time and geospatial resolution, so we had to find a way to get both datasets to join both datasets into cell-days for our model to use. We simply added all of the ACLED events for each cell-day and joined them to the cell-day as a sum called *Violence*. We then used the following metric score to calculate the impact of GDELT articles for a given cell-day, where *article_count* is the number of articles in that cell-day, the Goldstein score is a number between -10 and 10 indicating the article's contents' potential effect on the country's stability, and the tone is a sentiment value derived from natural language processing (negative values indicate more conflict):

$$\textbf{impact_score} = \ln(1 + \textit{article_count}) * \max(0, -\textit{avg_goldstein} * \textit{avg_tone})$$

We ascertained our model's bias toward zero in predicting violence. It optimized quickly for "silence" or a Violence value of 0 for almost all of the days in our rolling 15-day window. Due to the spontaneous and infrequent nature of the Thailand-Cambodia conflict, our model predicted that the vast majority of days would not have violence. Since we performed our validation testing with the set of anomalous days (those containing a much higher amount of tension starting in February 2025, plus the actual border clashes from May-July), the error score actually served as a well-functioning metric of tension. The model predicted non-violence for days with violence, leading to a high error and providing early warning of the escalating conflict.

One metric we were interested in seeing was whether our iterations on our foundational model produced any gains in predicting escalation, i.e. as the conflict ramped up, would our model be able to keep up and accurately predict the next time series, or would it continue to predict very low or the same values of violence? Our attention models proved to yield higher

performance in maintaining accurate predictions of escalation over periods of longer conflict (such as the June-July battles)–probably due to their ability to adaptively change input weights to reflect the normal state–but also higher overall performance in predicting sparse, infrequent periods of conflict. We suspect this is due to the temporal attention mechanism's ability to select relevant windows of time from across the range of inputs.

Discussion

Precise data is a crucial part for conflict forecasting and modeling. A problem we ran into when testing was the tendency for GDELT data to generalize its geolocation data toward administrative region centroids, meaning depending on the size of the designated region, our geolocation of a specific discrete event will be inaccurate and thus lead to higher violence and impact scores for a given cell-day.

We notice several trends present in the literature. First, low-intensity conflict is very hard to predict, with our unsupervised model exhibiting a bias towards zero due to extended periods of peace; we note that perhaps training our model on the anomalies may have yielded better results specifically for the Thailand-Cambodia border conflict, but would probably have been less transferrable to other regions. As Chadefaux (2017) observes, the goal of conflict study should be to yield methods that can be applied across areas and disciplines. As he also notes, peace brings peace and conflict invites more conflict; war can be started for completely irrational and unpredictable reasons. Our model is able to capture the underlying trends of tension and conflict from the event data, but it does not produce explanations for escalation and its predictions may not hold true across longer periods of time.

As we discuss in the next section, alternative or additional data streams may provide quicker indications of impending conflict activity. However, this does not mean our model is deprecated, as it can provide highly specialized spatiotemporal pattern recognition capabilities to support human analysis. For example, a humanitarian organization with limited resources may be able to allocate staff to a disputed area ahead of a conflict's outbreak using the predictions of a similar forecasting model. In this way, we believe our model is a complementary tool for domain experts and policymakers to use to avoid being blindsided by sudden conflict.

Suggestions

Several opportunities exist for extending the analytical framework presented in this study. While our model demonstrates potential for identifying escalation risk, additional data sources and methodological refinements could improve the accuracy and timeliness of our implementation for conflict early-warning.

One area for future research is the integration of physical terrain attributes into our predictive framework. Our current model primarily focuses on spatiotemporal conflict patterns and information signals, but does not explicitly incorporate physical characteristics of the disputed border areas. Terrain ruggedness, forest density, elevation changes, and transportation accessibility can significantly influence the likelihood of militarized encounters, particularly in regions where troop movement and artillery placement depend heavily on geography. These attributes are more suited to a geospatial approach through terrain analysis or remotely sensed data, including imagery methods that were used by Thailand to document fortifications and artillery positions ahead of the 2025 clashes (Nation Thailand, 2025). Incorporating GIS datasets

or imagery could allow models to better examine structural features that affect escalation in the disputed zones.

Another promising direction involves expanding the role of near-real-time data sources. An event's appearance in datasets such as ACLED and GDELT lags behind its occurrence due to those datasets' collection, verification, and reporting pipelines. Social media feeds, satellite imagery pipelines, and other open-source intelligence streams may provide earlier signals of emerging tensions. For example, sudden increases in localized social media activity, viral dissemination of battlefield footage, or observable changes in military infrastructure detected through remote sensing could serve as early indicators of escalation before formal reports are published. The challenge of integrating these streams is filtering noisy, low-quality elements, notwithstanding the effects of disinformation, propaganda, and influence operations in shaping the media space.

Using social media would almost certainly require a custom pipeline for scraping, filtering, and feature extraction. One solution that seemed promising was the use of a machine coding system like TABARI, which can take newswires and news articles and turn them into data points with actor and action fields similar to ACLED and GDELT (Best et al. 2013). This method balances data quality with speed of collection, allowing for more instantaneous model updates. A human coding pipeline could also be used for lower-trust sources and points requiring analysts with domain knowledge (such as geolocatable footage).

Future iterations of our model could also explore how different grid resolutions or spatial partitions influence model performance. We started off with the idea to use Uber's H3 hexagonal grid system to improve spatial scalability for aggregation and visualization, but abandoned this idea as it would add unnecessary complexity to our already burdened data pipeline. Integrating

dynamic spatial clustering methods may improve the identification of emerging hotspots that do not align neatly with predefined grid boundaries.

Another area for development involves combining anomaly detection with a supervised forecasting model. The current approach focuses on identifying deviations from historically normal patterns rather than predicting specific conflict outcomes. Hybrid frameworks that integrate anomaly detection signals with probabilistic forecasting models may help bridge this gap by translating detected anomalies into more interpretable escalation risk scores for policymakers and analysts.

Future work may benefit from incorporating additional contextual variables related to political developments and diplomatic signaling. Leadership changes, election cycles, bilateral negotiations, and shifts in defense policy can all influence the likelihood of escalation in territorial disputes. Integrating structured political event data alongside conflict and media signals could help models contextualize the broader strategic environment in which localized incidents occur.

Together, these avenues for future research highlight the broader potential of combining machine learning, geospatial analysis, and open-source intelligence to improve early warning capabilities in interstate conflicts. While predictive models cannot fully anticipate the complex political dynamics that drive escalation, continued methodological development may enhance the ability of analysts and policymakers to monitor emerging risks and respond proactively to evolving conflict conditions.

References

Best, R., Carpino, C., and Crescenzi, M. (2013) An analysis of the TABARI coding system. *Conflict Management and Peace Science,* 30 (4): 335-348.

Chadefaux, T. (2017) Conflict Forecasting and its Limits. *Data Science,* 1 (1-2): 7-17.

Chalermpalanupap, T. (2025) Ominous Political Storm Gathers Over Thai-Cambodian Dispute in Gulf of Thailand. ISEAS: *Yusof Ishak Institute,* 2025 (6): 3-8.

Hegre, H., Allansson, M., Basedau, M., Colaresi, M., Croicu, M., et al. (2019) ViEWS: A Political Violence Early-Warning System. *Journal of Peace Research,* 56 (2): 155–74.

Hochreiter, S. and Schmidhuber, J. (1997) Long Short-Term Memory. *Neural Computation*, 9 (8): 1735–1780.

Macis, L., Tagliapietra, M., Meo, R., and Pisano P. (2024) Breaking the Trend: Anomaly Detection Models for Early Warning of Socio-political Unrest. *Technological Forecasting and Social Change,* 206 (6): 123459.

Nation Thailand. "Thai Military Releases Aerial Evidence of Cambodian Military Buildup before Border Clash", 10 June 2025. https://www.nationthailand.com/news/general/40051042.

Qin, Y., Song, D., Chen, H., Cheng, W., Jiang, G., Cottrell, G. (2017) A Dual-Stage Attention-Based Recurrent Neural Network for Time Series Prediction. *IJCAI'17: Proceedings of the 26th International Joint Conference on Artificial Intelligence*, 2627-2633.

Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010) Introducing ACLED: An Armed Conflict Location And Event Dataset. *Journal of Peace Research,* 47 (5): 651–660.

Schrodt, P. A., Yonamine, J., and Bagozzi, B. E. (2013) Data-Based Computational Approaches To Forecasting Political Violence. *Handbook of Computational Approaches to Counterterrorism*, 129–162.

Tollefsen, A. F., Strand, H., & Buhaug, H. (2012) PRIO-GRID: A Unified Spatial Data Structure. *Journal of Peace Research*, 49 (2): 363-374.

Wagener, M. (2011) Lessons from Preah Vihear: Thailand, Cambodia, and the Nature of Low-Intensity Border Conflicts. *Journal of Current Southeast Asian Affairs*, 30 (3): 27-59.

Yuxuan, L., Songyu, K., Junbo, Z., Xiuwen, Y., and Yu, Z. (2018) GeoMAN: multi-level attention networks for geo-sensory time series prediction. *IJCAI'18: Proceedings of the 27th International Joint Conference on Artificial Intelligence*, 3428–3434.

No original data was used in the production of this paper. Country shapefiles were obtained from https://gadm.org/download_country.html. PRIO-GRID shapefiles were obtained from https://viewsforecasting.org/gis-resources/. SIPRI: https://www.sipri.org/databases/milex.